Futures Time-Series Momentum

ORIE 5260 · Midterm Project Presentation

Data validation, strategy construction, attribution, and risk forecasting

58 futures markets · Data: 1969–2014

Main evaluation: 1976–2014

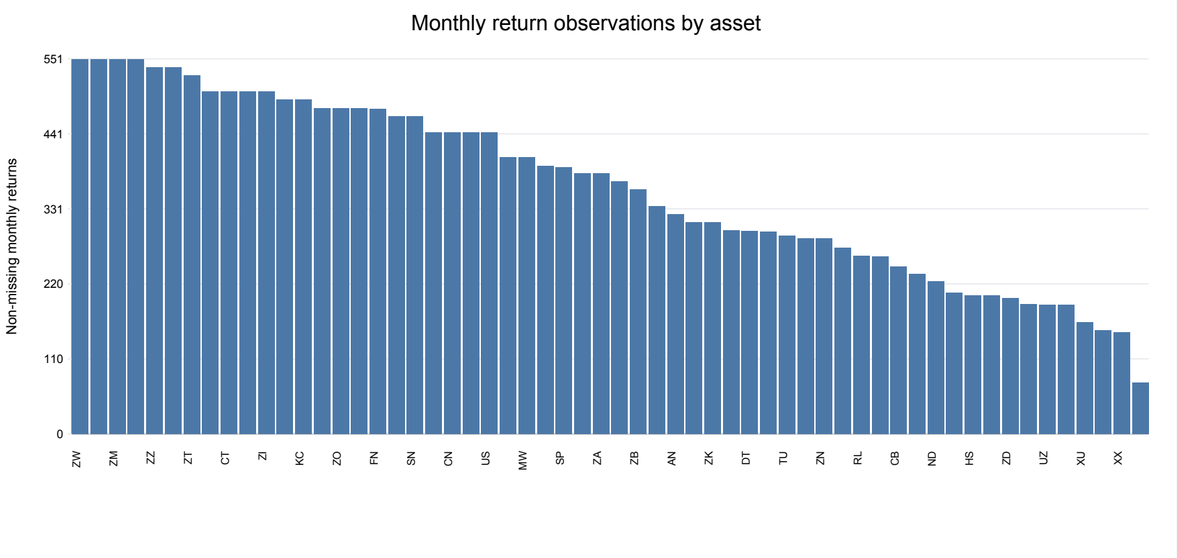
Based on midterm_report.md, project code, and existing outputs

Research question and completed work

Does past return direction predict next-month futures returns?

- 58 markets across commodities, equity indices, fixed income, and FX.
- Validate data, construct the course PPT strategy, and test four signal horizons.
- Evaluate performance, return sources, risk forecasts, and SG Trend.
- Compare completed enhancements and identify the next model improvements.

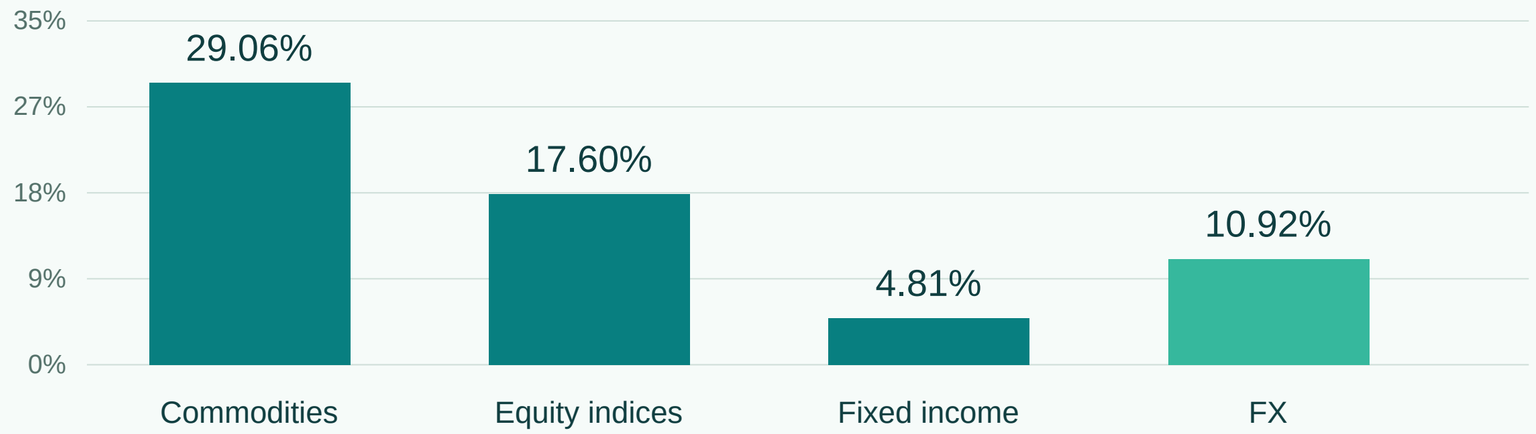
Coverage: the universe changes over time



- 20,987 valid observations out of 32,016 possible cells.
- 11,029 missing cells: 34.45% of the full panel.
- ZW has 551 observations; AP has 75.
- Different histories change both eligible market counts and historical comparisons.

outputs/eda_output/missingness.csv; observations_by_asset.svg

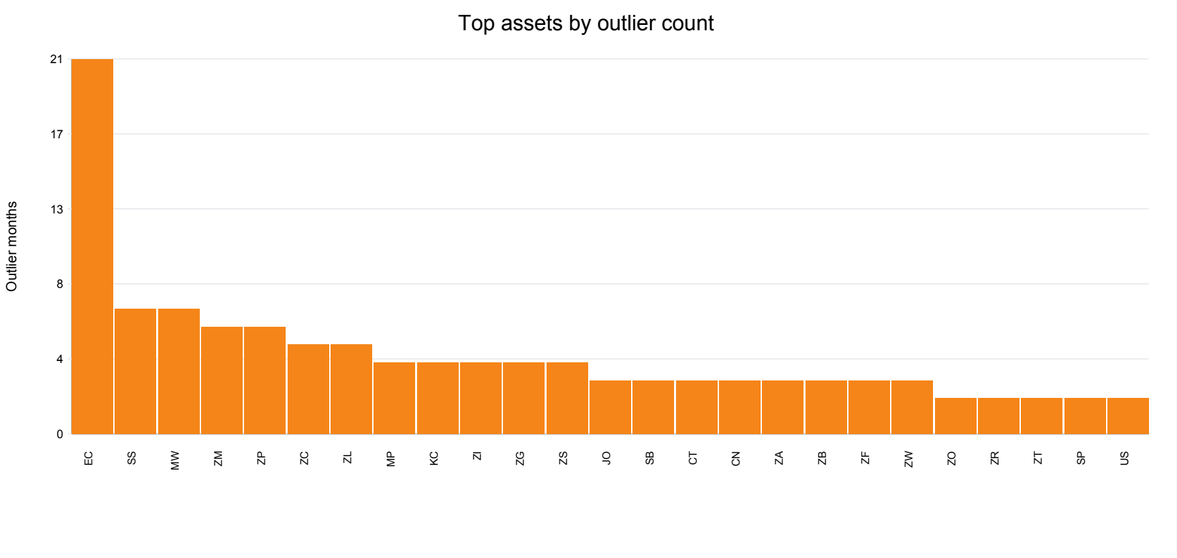
Different market risks motivate volatility scaling



Commodity median volatility is about six times fixed income volatility. Equal dollar weights would give unequal standalone risk.

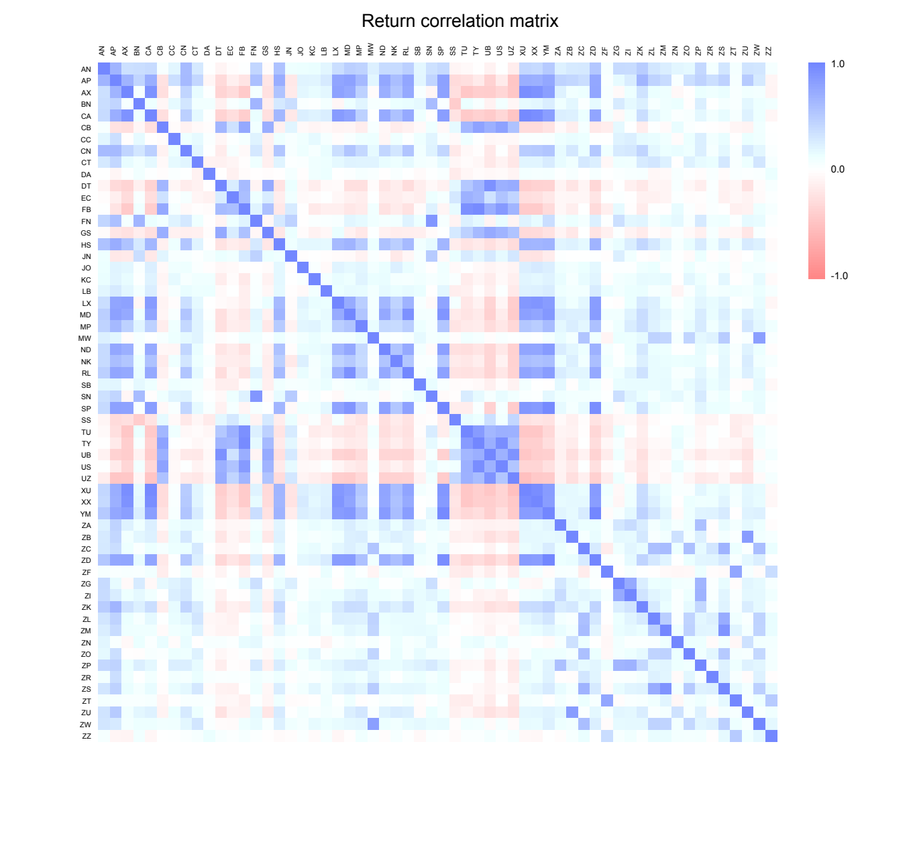
summary_statistics.csv; medians across instrument-level annualized volatility

Data quality: flags, reconstruction, and open checks



- 126 outliers flagged with 3×IQR or |robust z| > 5; retained for review.
- 14,974 daily-to-monthly overlapping checks; typical differences approximately zero.
- RL maximum discrepancy: 4.70 percentage points, unresolved.
- No history backfill; missingness includes different start and end dates.

Cross-market correlation challenges independence



Class	Mean within-class correlation
Commodities	0.15
Equity indices	0.76
Fixed income	0.62
FX	0.28

Equity and fixed-income markets move together strongly. Contract-level breadth can overstate diversification.

EDA implications for strategy construction

Data finding	Design response	Remaining limitation
Unequal market histories	Dynamic eligibility and at least 36 valid months	Composition changes over time
Large differences in volatility	Inverse-volatility instrument scaling	No explicit portfolio risk target
Heavy tails and extreme observations	Retain flags; report drawdowns and robustness	Outlier sensitivity remains future work
Strong related-market correlations	Diagnose concentration and validate diagonal forecasts	Covariance omitted from current baseline
RL reconstruction exception	Keep discrepancy visible	Resolve before final conclusions

Two baseline versions: rolling and course PPT

Dimension	Rolling baseline	PPT / adjusted baseline
Script	rolling_backtest.py	strategy_backtest_adj.py
12-month signal	sign[∏(1+r) − 1]	sign[Σr]
Volatility estimate	Trailing 12 months	Expanding; at least 36 valid months
Evaluation sample	Feb 1970–Dec 2014; 539 months	Jan 1976–Dec 2014; 468 months
TSMOM CAGR / Sharpe	19.12% / 1.06	10.64% / 0.97
TSMOM vol / MaxDD	18.05% / −27.58%	11.15% / −26.03%

Both use historical inputs, execute in the next month, and compound wealth. Different signals, risk windows, eligibility, and samples prevent a signal-only comparison.

rolling_backtest_output/performance_summary.csv: rolling_tsmom, 0 bps; strategy_backtest_adj_output/baseline_summary.csv

The course PPT approach: adjusted baseline

$$s_{i,t} = \text{sign}\left(\sum_{k=0}^{11} r_{i,t-k}\right)$$

$$\hat{\sigma}_{i,t} = \text{Std}(r_{i,1:t}) \times \sqrt{12}$$

$$w_{i,t} = \frac{1}{N_t} \times s_{i,t} \times \frac{0.40}{\hat{\sigma}_{i,t}}$$

- Expanding volatility requires at least 36 valid monthly observations.
- Weights formed in month t earn returns in t+1; rebalance monthly.
- Static uses the same volatility and eligibility, always long.

strategy_backtest_adj.py follows the course PPT approach. The 40% parameter scales individual markets; it is not a portfolio volatility target.

Execution timing and assumptions

Formation month t	Execution month t+1
Signal and risk estimates use data through t	execution_weights = weights.shift(1)
Construct equally allocated inverse-volatility positions	Rp,t□1 = Σ wi,t ri,t□1
Keep formation and execution alignment separately	Timing checks confirm a one-month lag

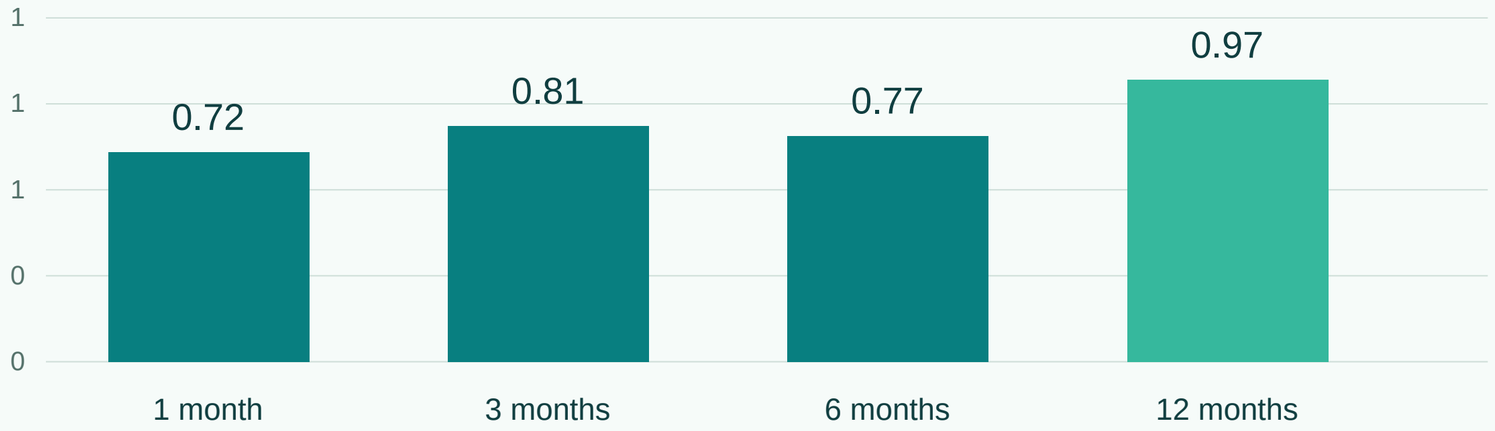
- Missing realized returns for held positions are filled with zero.
- No leverage cap, margin constraints, financing, or explicit roll costs.
- The diagonal risk model assumes zero cross-market covariance.

Momentum signals contain predictive information

Lookback	Positive-minus-negative spread	Directional hit rate	Vol-scaled Sharpe	HAC t
1 month	0.58%	52.8%	0.57	3.61
3 months	0.79%	53.5%	0.77	4.10
6 months	0.72%	54.0%	0.72	3.73
12 months	0.79%	54.4%	0.86	4.51

The 12-month signal has the highest hit rate and scaled Sharpe. The 3-month signal is also strong, motivating multiple horizons.

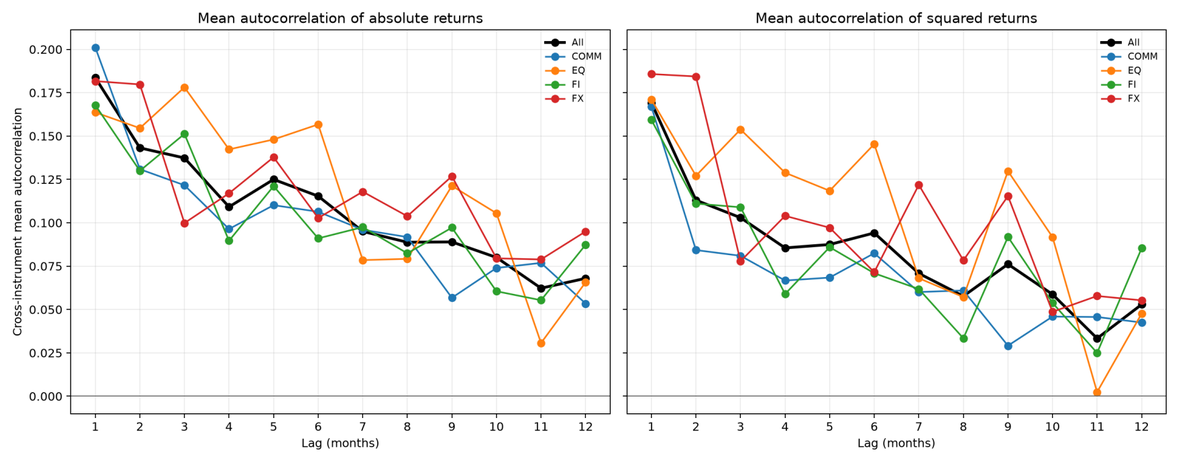
Four lookbacks: complete portfolio backtests



Lookback	1 month	3 months	6 months	12 months
CAGR	7.71%	9.23%	8.47%	10.64%
Annual vol	11.23%	11.81%	11.36%	11.15%

strategy_backtest_adj_output/lookback_summary.csv; 1976–2014; RF = 0

Volatility persists more than raw returns



18.4%

Mean lag-1 autocorrelation of absolute returns

6.2%

Mean lag-1 autocorrelation of raw returns

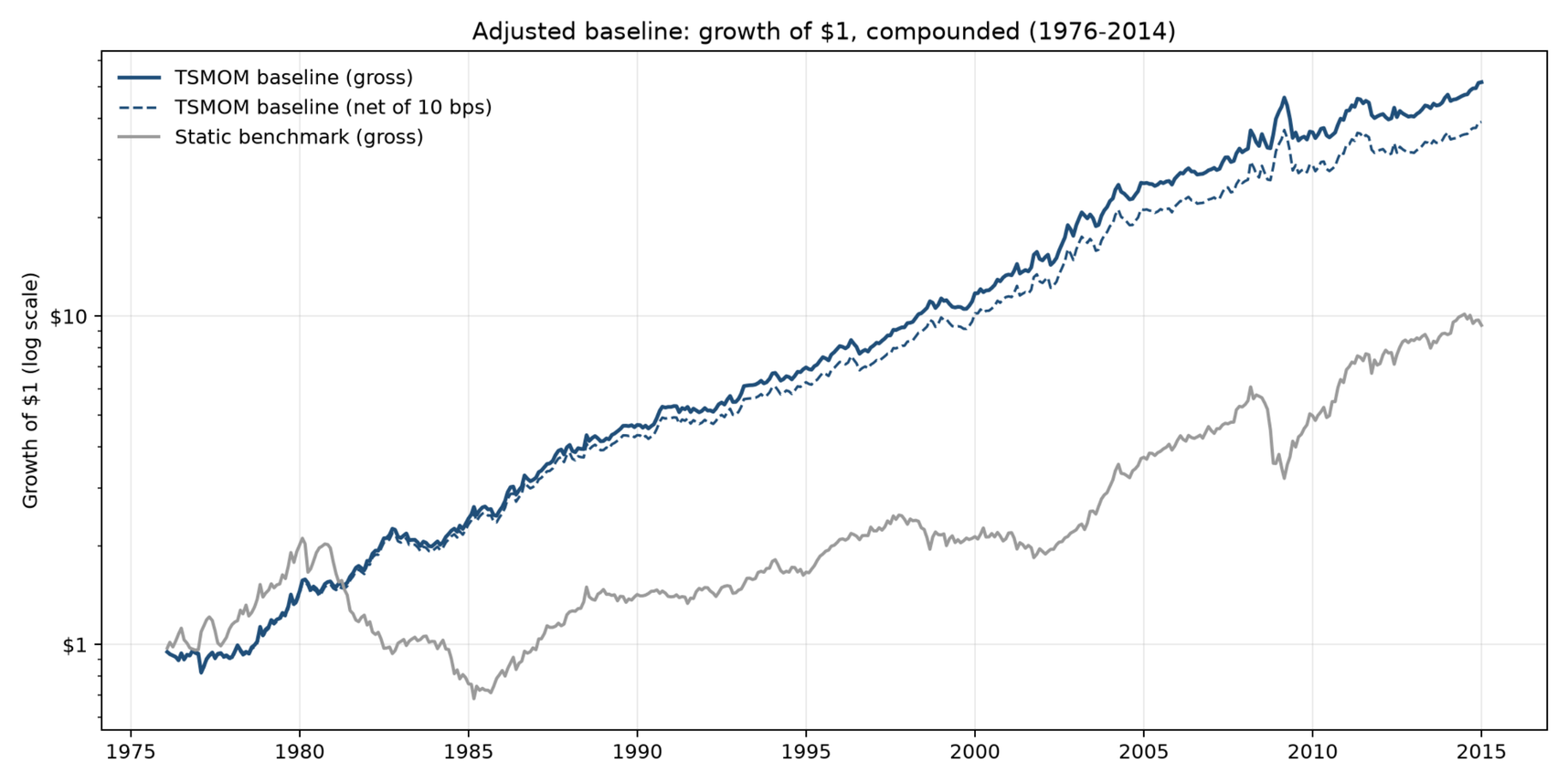
Volatility clustering supports dynamic risk estimates. Weak raw-return autocorrelation makes direct signal testing necessary.

Main results: stronger returns and smaller drawdowns

1976–2014; RF = 0	PPT / adjusted TSMOM	Static long
CAGR	10.64%	5.91%
Annualized mean	10.77%	6.82%
Annualized volatility	11.15%	14.52%
Sharpe: annualized mean / vol	0.97	0.47
Maximum drawdown	−26.03%	−67.54%
Positive-return months	64.1%	60.0%
Annual turnover	7.24	0.41

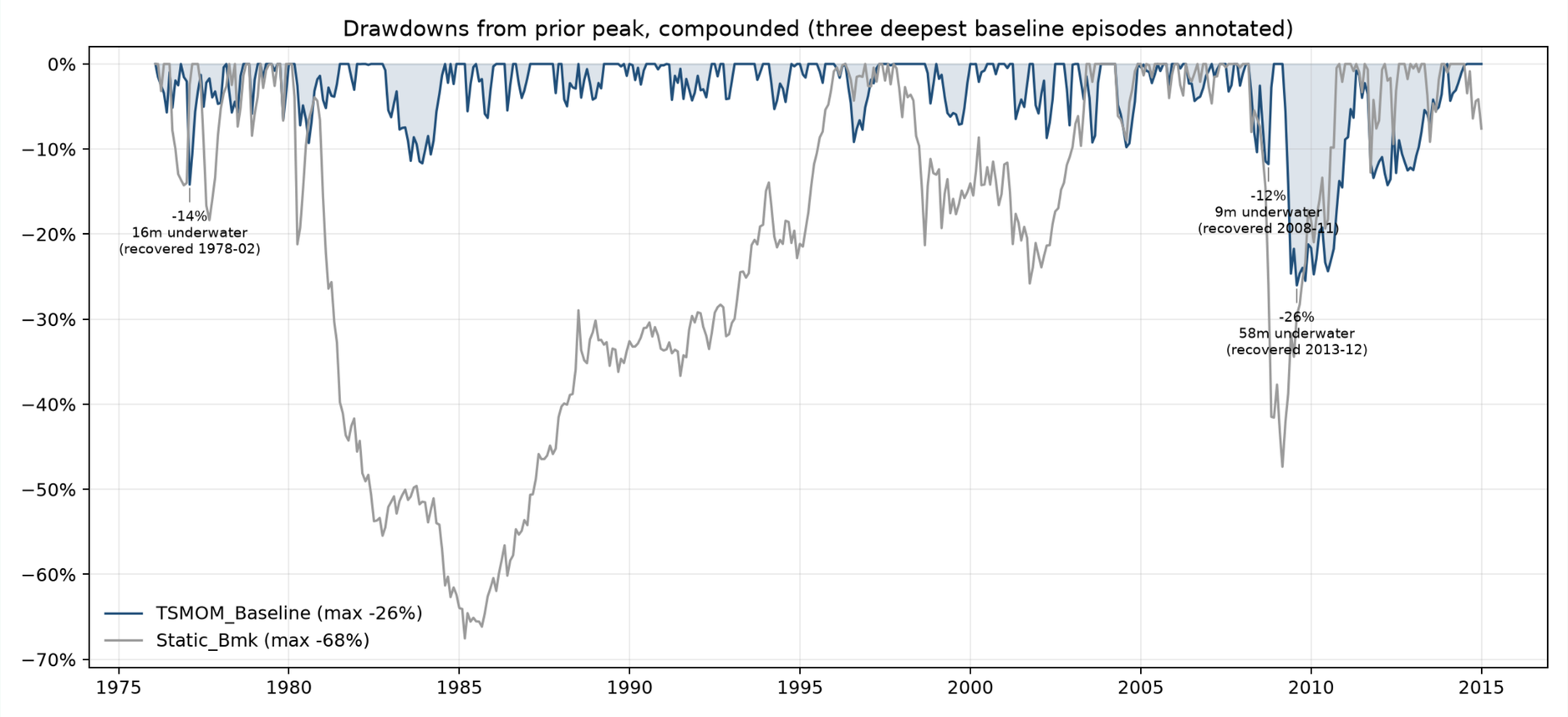
baseline_performance_summary.csv□strategy_backtest_adj_output/baseline_summary.csv

Compounded wealth of the main strategy



outputs/baseline_performance_output/baseline_cumulative_return.png

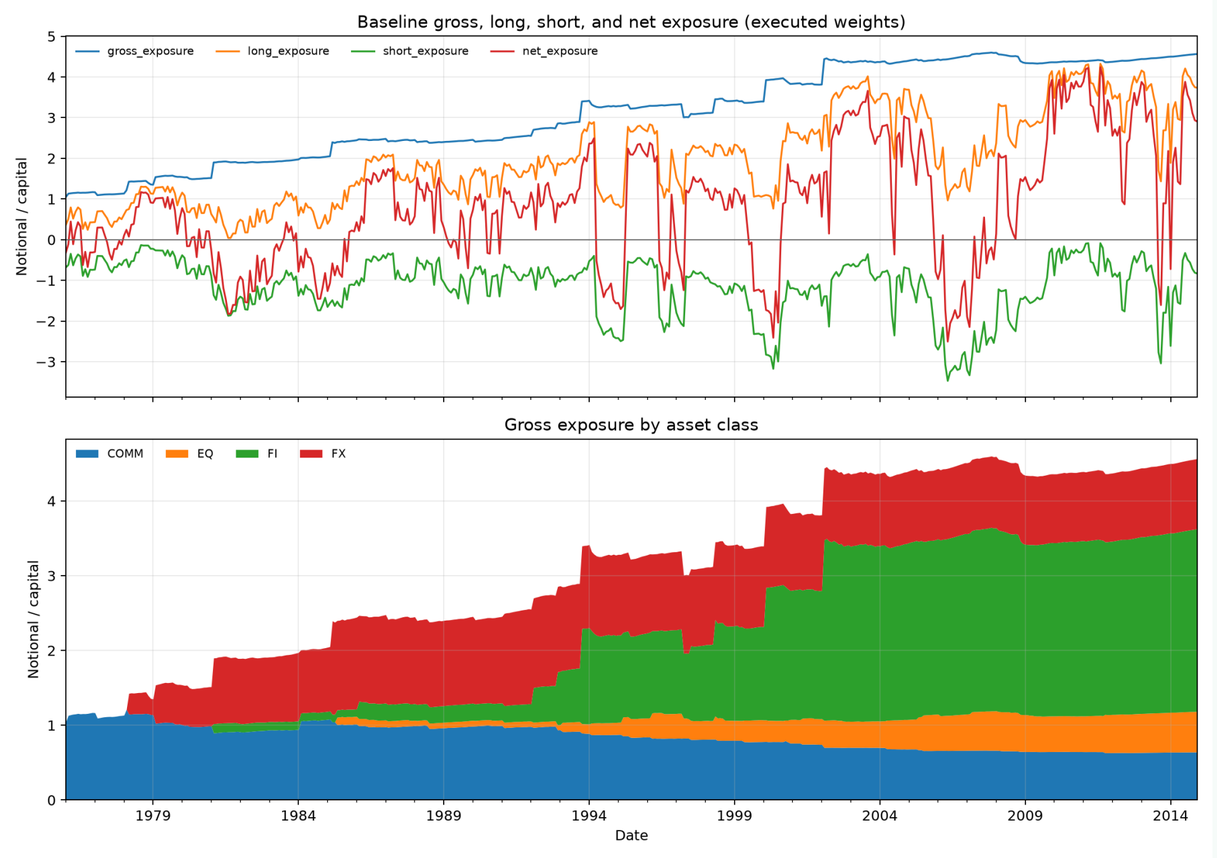
Drawdowns: the cost of staying invested



TSMOM maximum drawdown: −26.03%; static: −67.54%.

outputs/baseline_performance_output/baseline_drawdowns.png

Exposure: implicit leverage and average long bias



Average gross exposure: 3.13; average net exposure: +0.84.

outputs/baseline_performance_output/baseline_exposures.png

Rolling performance varies across periods



2010–2014 annualized mean return: 7.57%, below earlier decades.

outputs/baseline_performance_output/baseline_rolling_performance.png

Transaction costs and turnover

Cost per one-way weight change	CAGR	Sharpe	Annual return drag
0 bps	10.64%	0.97	0.00%
5 bps	10.24%	0.93	0.36%
10 bps	9.85%	0.90	0.72%
25 bps	8.68%	0.81	1.81%

- Apply costs to monthly Σ|Δw|; TSMOM trades much more than static.
- Profitability remains positive at all tested costs; Sharpe is 0.81 at 25 bps.
- Contract-specific spreads, roll, financing, liquidity, and impact are omitted.

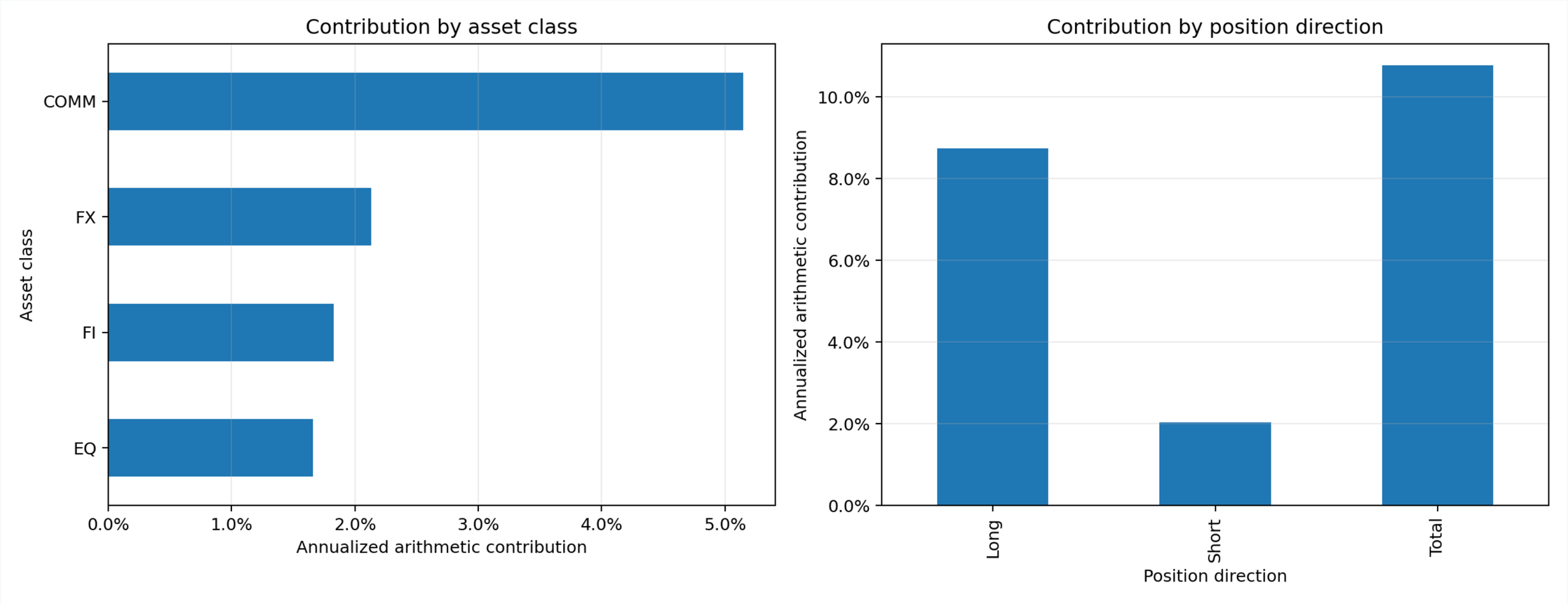
outputs/baseline_performance_output/transaction_cost_sensitivity.csv

Return sources: every asset class contributes

Source	Annualized mean contribution	Share of total
Commodities	5.15%	47.8%
FX	2.13%	19.8%
Fixed income	1.83%	17.0%
Equity indices	1.66%	15.4%
Long positions	8.74%	81.2%
Short positions	2.03%	18.8%

- Returns are broad, but commodities and long positions dominate.
- Lumber and gold are the largest market contributors.
- Soybeans and orange juice detract most.
- Instrument contributions reconcile to portfolio arithmetic returns.

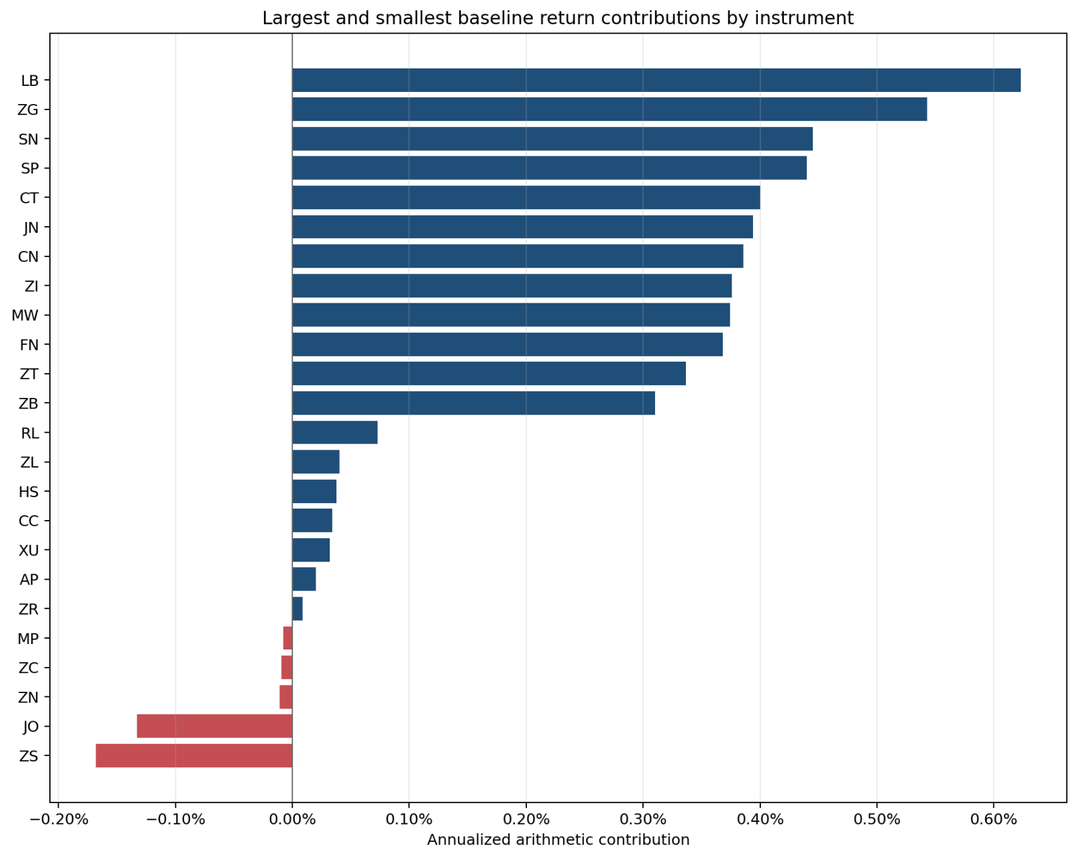
Asset classes and trading direction generate returns



All classes and both directions contribute; commodities and longs dominate.

outputs/baseline_return_attribution_output/asset_class_and_direction_contribution.png

Market-level attribution reveals concentration



Lumber and gold contribute most; soybeans and orange juice detract most.

outputs/baseline_return_attribution_output/instrument_contribution.png

Return attribution by regime and decade

Internal S&P futures regime	Annualized mean	Annualized volatility
Bear	10.26%	12.60%
Bull	11.13%	8.33%

Decade	1970s	1980s	1990s	2000s	2010–2014
Annualized mean	10.37%	12.22%	9.55%	12.29%	7.57%

Returns are positive in every reported decade and both regimes. Bear-regime risk is higher, and recent returns are weaker.

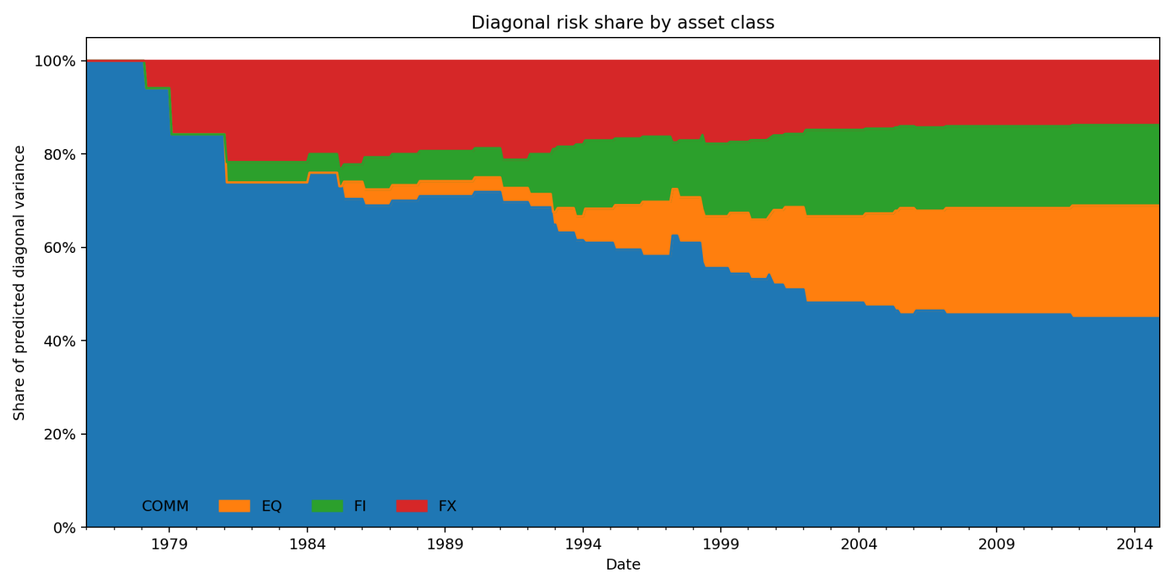
portfolio_return_by_regime.csv portfolio_return_by_decade.csv

Six-factor regression: significant momentum exposure

Joint regression, 2000–2014	Alpha / Beta	HAC t	p
Annualized alpha	7.20%	2.27	0.023
Mkt-RF	0.004	0.05	0.964
SMB	−0.005	−0.06	0.949
HML	−0.222	−1.71	0.087
Mom	0.286	5.17	<0.001
RMW	0.184	1.76	0.078
CMA	0.218	1.31	0.189

R^2 = 28.0%. Equity factors do not fully explain futures returns; residual alpha is not automatically evidence of manager skill.

Many markets, concentrated asset-class risk



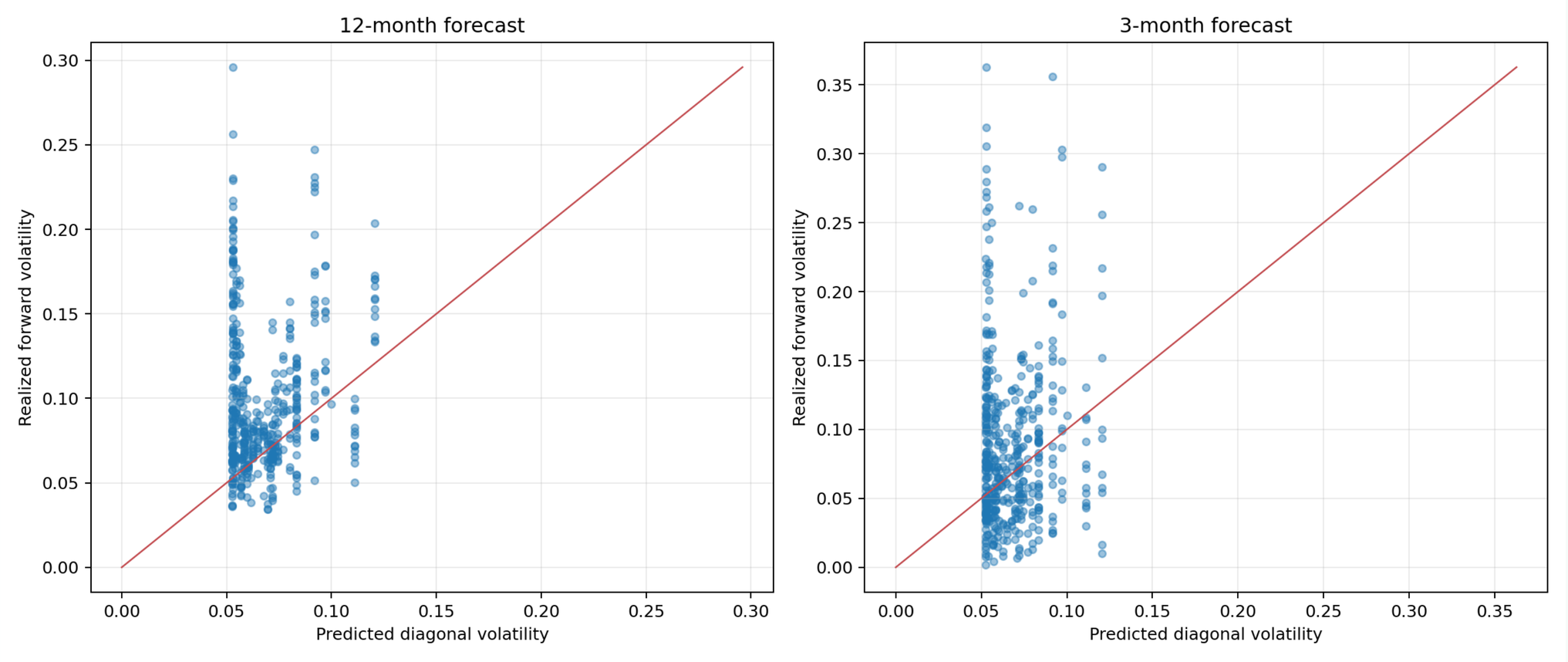
- Commodities: 62.7% of average predicted variance.
- FX: 15.9%; fixed income: 11.0%; equities: 10.4%.
- Effective instrument risk positions: 39.6.
- Effective asset-class risk positions: 2.35.
- A large market count does not ensure diversified common risks.

The diagonal model understates future risk

Fixed-weight horizon	Predicted vol	Realized vol	Realized − predicted	Correlation	Realized exceeds forecast
Next 12 months	6.82%	9.78%	+2.96pp	0.13	74.8%
Next 3 months	6.79%	8.87%	+2.08pp	0.08	54.6%

- Freeze formation-date weights and estimate volatility from subsequent returns.
- At 12 months, realized risk exceeds the forecast by more than 25% in 50.4% of observations.
- Ignoring correlation among same-direction positions is a plausible source of bias.

Weak linear risk-forecasting power



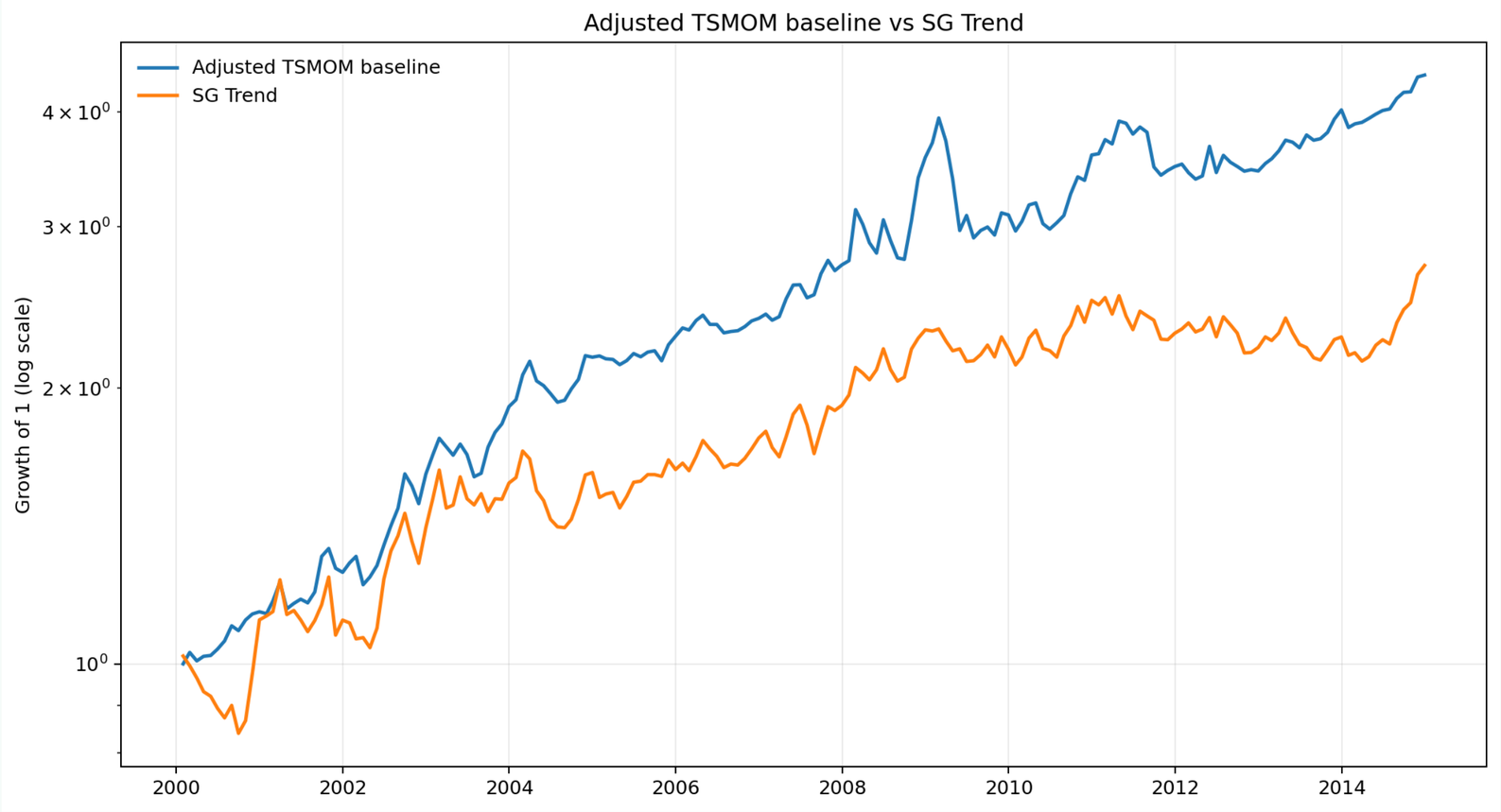
outputs/risk_model_validation_output/forecast_scatter.png

SG Trend: shared trend exposure, different drawdowns

Common sample: 2000–2014	PPT / adjusted TSMOM	SG Trend
Annualized mean	10.72%	7.75%
CAGR	10.37%	6.90%
Annualized volatility	12.83%	14.60%
Sharpe; RF = 0	0.84	0.53
Maximum drawdown	−26.03%	−17.66%

Correlation: 0.62. Raw-return regression beta: 0.54; annualized alpha: 6.52%; HAC t: 2.48.

Compounded performance versus SG Trend



outputs/strategy_analysis_output/sg_trend_cumulative.png

Rolling baseline: performance and cost sensitivity

Rolling baseline, Feb 1970–Dec 2014	CAGR	Annual vol	Sharpe	MaxDD
TSMOM, 0 bps	19.12%	18.05%	1.06	−27.58%
TSMOM, 5 bps	18.21%	18.05%	1.02	−27.76%
TSMOM, 10 bps	17.30%	18.06%	0.97	−27.93%
TSMOM, 25 bps	14.60%	18.12%	0.84	−28.45%
Static, 0 bps	13.33%	21.89%	0.68	−71.78%

- Compounded 12-month signal with trailing 12-month volatility.
- TSMOM and static share the same eligible universe.
- Average monthly traded notional: 1.30; compare costs separately from the PPT baseline.

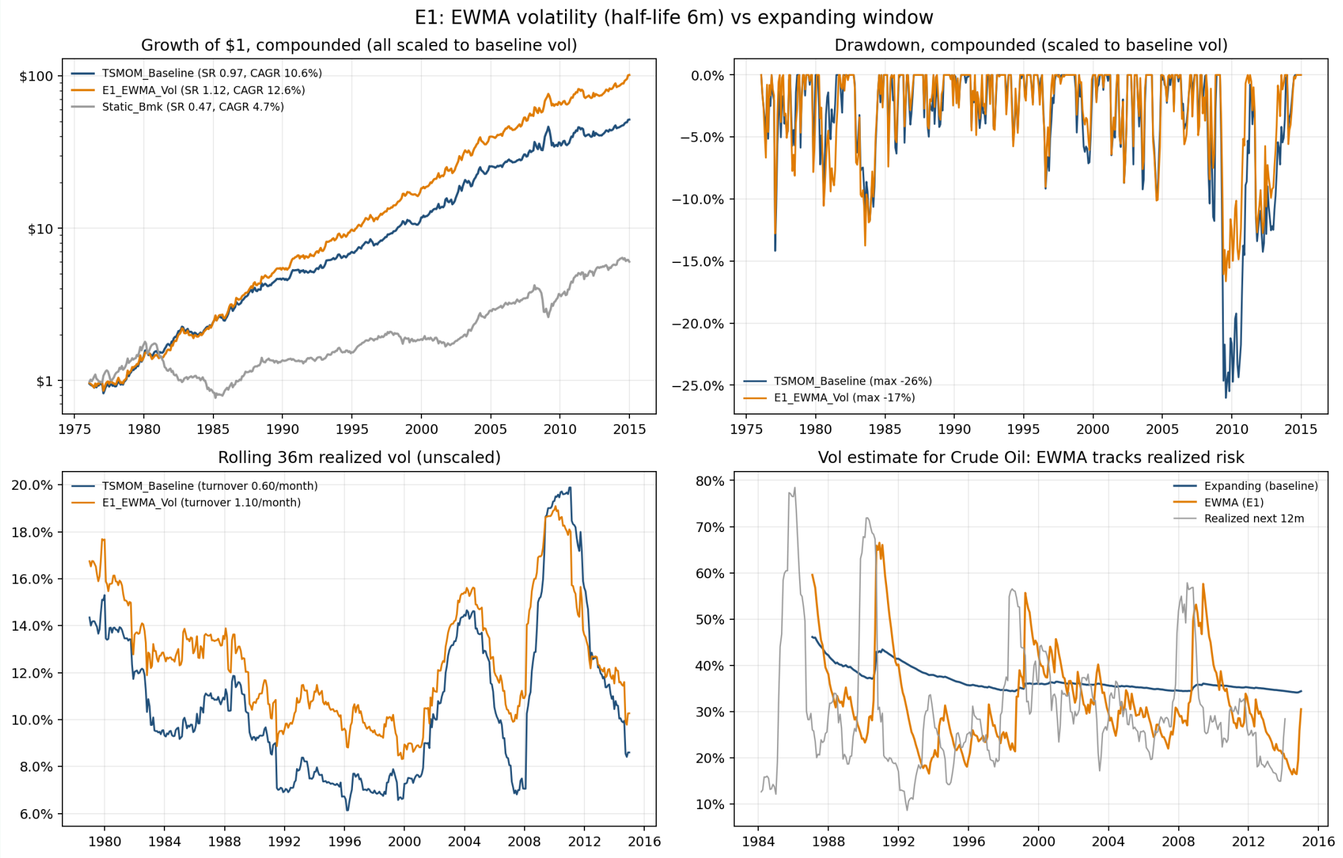
rolling_backtest.py; rolling_backtest_output/performance_summary.csv: rolling_tsmom / rolling_static rows only

Four completed PPT / adjusted experiments

Experiment	Change from baseline	Purpose
E1: EWMA volatility	6-month half-life; keep 36-month eligibility	Respond faster to changing risk
E2: Asset-class budget	Equal allocation across valid classes; 1/N within class	Reduce market-count imbalance
E3: Multiple lookbacks	Average 1-, 3-, 6-, and 12-month signs	Reduce dependence on one horizon
E4: Combined	E1 + E2 + E3	Test whether improvements combine

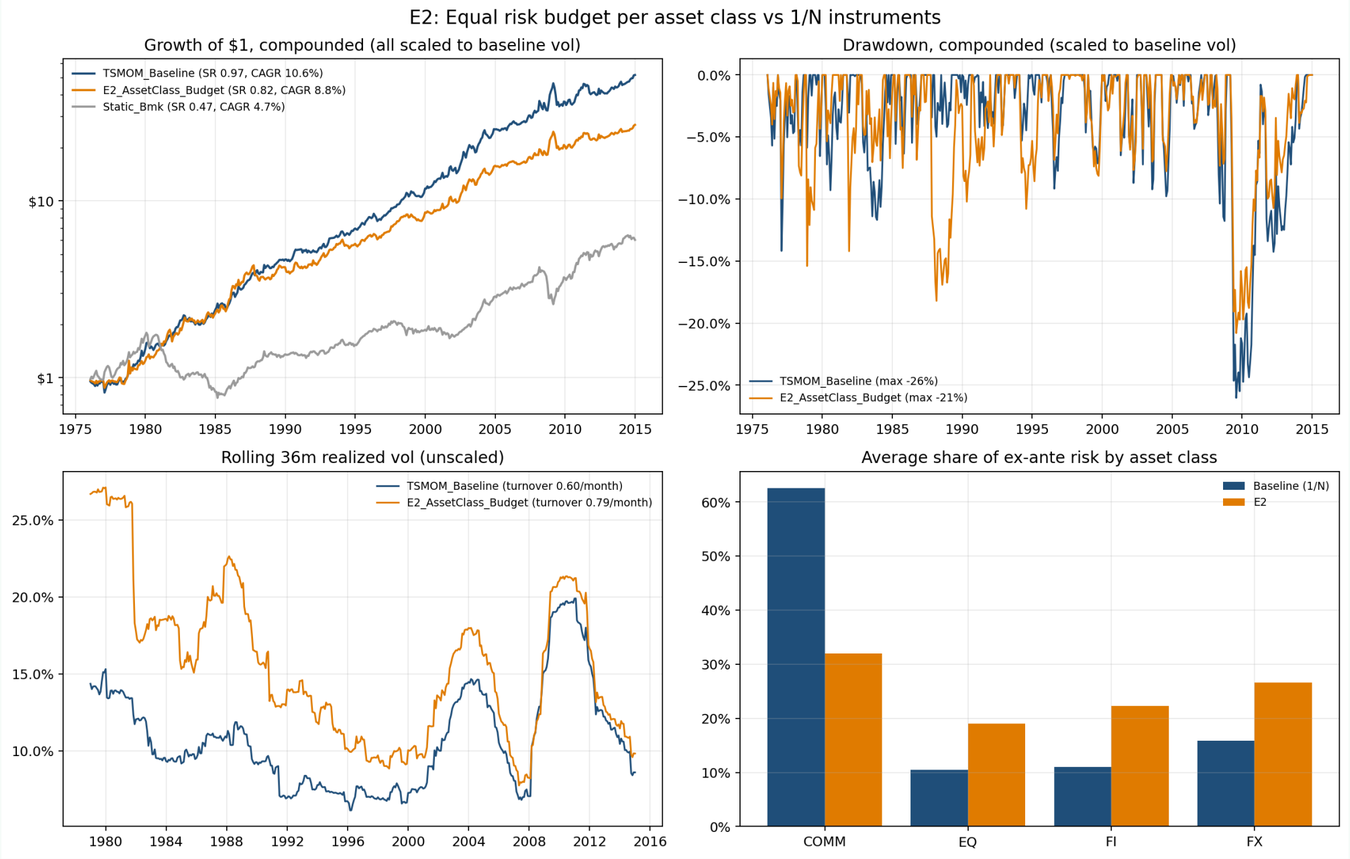
These experiments have results. Full covariance models, portfolio volatility targeting, and walk-forward evaluation remain next-stage work.

E1: faster volatility estimates



EWMA half-life: 6 months. Gross Sharpe: 1.12; turnover increases.

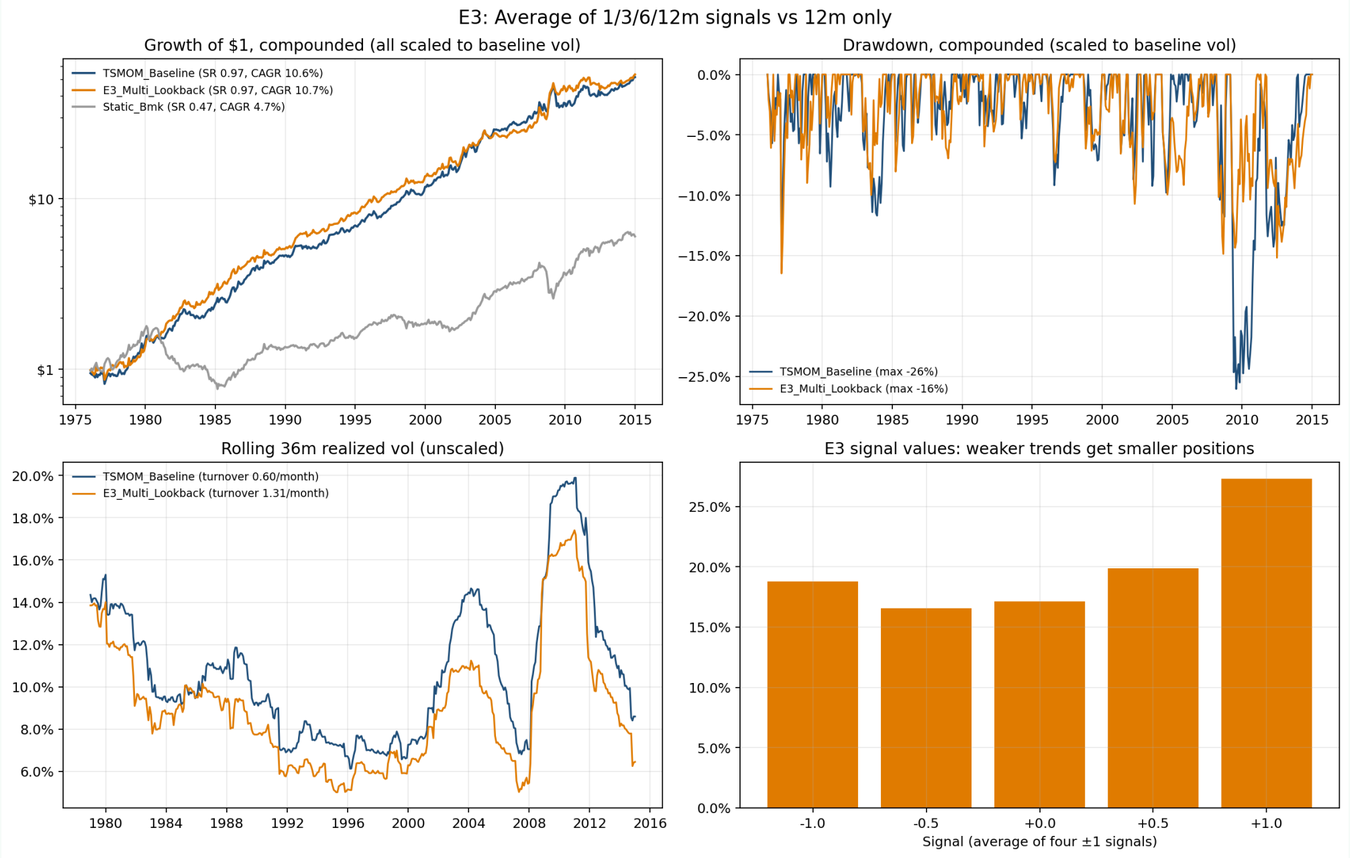
E2: allocation across asset classes



Equal class allocation changes concentration, but gross Sharpe falls to 0.82.

outputs/strategy_backtest_adj_output/experiment_E2.png

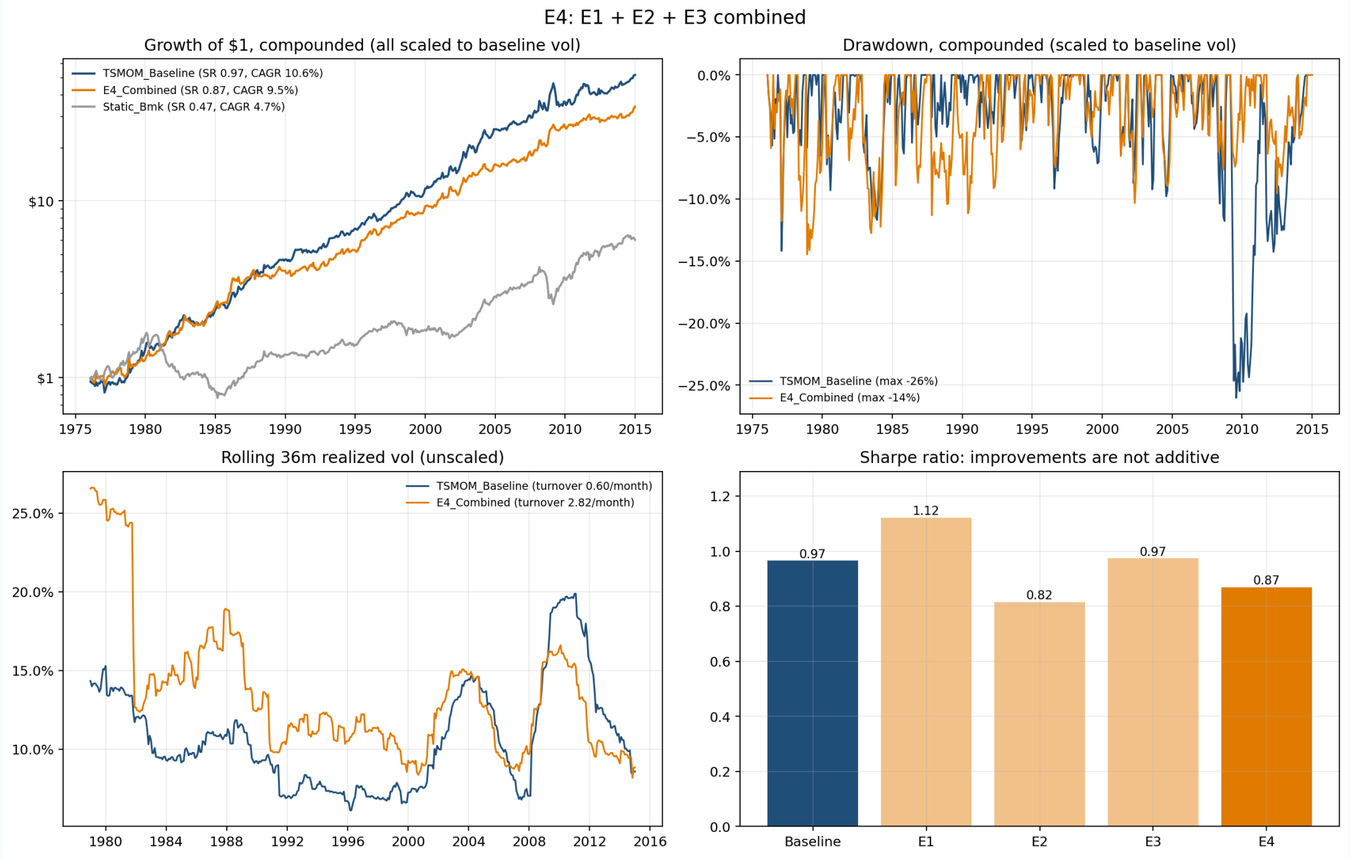
E3: combine four signal horizons



Average 1-, 3-, 6-, and 12-month signs; Sharpe 0.97 with smaller drawdowns.

outputs/strategy_backtest_adj_output/experiment_E3.png

E4: combined changes are not automatically better



EWMA + class allocation + multiple horizons: Sharpe 0.87; highest turnover.

Experimental improvements do not simply add up

Strategy	Raw Sharpe	Matched-vol CAGR	Matched-vol MaxDD	Monthly turnover
Baseline	0.97	10.64%	-26.03%	0.60
E1_EWMA_Vol	1.12	12.57%	-16.63%	1.10
E2_AssetClass_Budget	0.82	8.81%	-20.81%	0.79
E3_Multi_Lookback	0.97	10.74%	-16.46%	1.31
E4_Combined	0.87	9.48%	-14.46%	2.82

- E1 reaches a 1.12 Sharpe, with substantially more turnover.
- E3 has a similar Sharpe to baseline and smaller drawdowns, with higher turnover.
- E2 and E4 have lower Sharpe; allocation and costs require further assessment.

strategy_backtest_adj_output/experiment_summary.csv

Findings: credible signal, incomplete risk model

- Predictiveness: the 12-month signal has a 54.4% hit rate and supportive HAC evidence.
- Historical performance: PPT / adjusted CAGR of 10.64% and Sharpe of 0.97.
- Return sources: all classes and both directions contribute; commodities and longs dominate.
- Risk gap: predicted volatility of 6.82% versus subsequent realized volatility of 9.78%.
- Implementation: leverage, turnover, zero-filled returns, and omitted costs matter.
- Experiments: EWMA improves results; multiple horizons reduce drawdowns; effects are not additive.

Next stage: covariance and implementable risk control

Priority	Plan and evaluation
Risk models	Expanding, rolling, EWMA, shrinkage covariance; compare fixed-weight forecast errors
Portfolio target	Ex-ante volatility scaling; leverage and scaling caps; volatility floors
Allocation and signals	Class-risk constraints and multiple horizons; separate risk-model and leverage effects
Implementation and validation	Realistic costs, roll, financing, liquidity; walk-forward with parameters fixed in advance
Data and factors	Review RL and outlier sensitivity; add carry, trend, duration, and USD factors

Reproducibility: two baselines, one attribution pipeline

Two baseline scripts have separate outputs. Detailed attribution and risk analysis use the standardized PPT / adjusted baseline.

Stage	Scripts / outputs
Data	analysis_eda.py; MonthlyReturns.csv; AssetMapCsv.csv
Rolling baseline	rolling_backtest.py; outputs/rolling_backtest_output/
PPT / adjusted baseline	strategy_backtest_adj.py; outputs/strategy_backtest_adj_output/
Standardized adjusted results	baseline_performance_analysis.py; baseline_output_loader.py
Attribution and validation	baseline_return_attribution.py; baseline_risk_analysis.py; validate_diagonal_risk_model.py
Signals and benchmarks	signal_diagnostics.py; generate_strategy_analysis.py

midterm_report.md; current code and baseline output tables